

❏ 1. Packet Filtering

Packet filtering is the **simplest and earliest form of firewall technology**. It inspects the **header information** of network packets and makes allow/block decisions based on predefined rules.

❏ Features:

- Works at OSI Layer 3 (Network) and Layer 4 (Transport)
- Filters packets by:
 - Source/Destination IP
 - Source/Destination Port
 - Protocol (TCP/UDP/ICMP)
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No inspection of packet **payload**

☒ **Pros:**

- Lightweight and fast
- Easy to configure
- Useful for basic access control

☒ **Cons:**

- Stateless — doesn't track connections
- Can't inspect data payload or application-level traffic
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Susceptible to spoofing and session hijacking

❏ 2. Screened Host Firewall

A **Screened Host Firewall** architecture is a **hybrid model** combining a **router (packet filter)** and a **bastion host** (hardened internal system).

❏ Structure:

csharp

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Internet | [Packet Filtering Router] | [Bastion Host (Firewall)] | Internal Network

❏ Key Concepts:

- **Router** filters general traffic (Layer 3/4)

- **Bastion Host** provides detailed inspection/logging (Layer 7)
- **Bastion Host** is the only system exposed to the external network

☒ **Pros:**

- More secure than standalone packet filter
- Allows application-level inspection via bastion
- Isolates internal network

☒ **Cons:**

- Bastion host can become a bottleneck

- Slightly more complex to configure and maintain
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☒ 3. Stateful Inspection Firewall

A **Stateful Inspection Firewall** (also called **Dynamic Packet Filtering**) maintains a **connection table** (**state table**) of active sessions.

☒ How It Works:

- Monitors **context** of packets, not just headers
- Remembers outgoing connections and allows related inbound packets
- Analyzes TCP handshakes and port numbers

☒ Characteristics:

- Operates at **Layers 3, 4, and partially 5**
- **Tracks state and sequence numbers**

☒ **Pros:**

- More secure than stateless packet filters
- Can block abnormal traffic flows (e.g., TCP without SYN)

☒ **Cons:**

- Heavier resource usage
- Doesn't inspect application payload (unless combined with DPI)

☒ 4. Next Generation Firewall (NGFW) – Application Controls

NGFWs go beyond stateful inspection by integrating deep packet inspection (DPI) and application-layer controls.

☒ Features of NGFW:

- Identifies applications regardless of port/protocol (e.g., Facebook, Skype)
- Provides URL filtering, malware detection, and IPS
- Supports identity-based rules (users/groups)
- Integrates with cloud threat intelligence

☒ Application Control Use Cases:

- Block specific app features (e.g., Facebook chat, but allow posts)
- Enforce bandwidth limits per app
- Detect encrypted traffic (SSL/TLS inspection)

☒ Pros:

- Extremely granular control
- Combines multiple security functions
- Supports Zero Trust and SASE models

☒ Cons:

- Expensive
 - Requires regular updates (app signatures)
 - May impact performance
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❏ 5. iptables – Linux Firewall Tool

iptables is the **built-in firewall** utility for **Linux systems**, based on the **netfilter** framework. It allows sysadmins to define rules for how packets should be handled.

❏ iptables Key Concepts:

Chain	Purpose
INPUT	For traffic destined to the local machine
OUTPUT	For traffic originating from the machine

FORWARD	For routed traffic (not local)
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Table	Use
filter	Default table, used for packet filtering
nat	For network address translation
mangle	For packet alterations (TTL, QoS)

☒ Sample iptables Rules:

bash

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```
# Block incoming SSH (port 22) sudo iptables -A INPUT -p tcp --dport 22 -j DROP # Allow HTTP traffic sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT # Set default policies sudo iptables -P INPUT DROP sudo iptables -P FORWARD DROP sudo iptables -P OUTPUT ACCEPT
```

☒ Pros:

- Highly configurable and powerful
- Native to Linux (no 3rd-party software needed)

☒ Cons:

- Complex syntax
- Hard to manage in large environments
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Replaced in many distros by nftables

☒ Summary Table

Firewall Type	Key Feature	OSI Layer(s)
Packet Filter	Filters IP/Port/Protocol	3, 4
Screened Host	Router + Bastion Host	3 to 7
Stateful Inspection	Tracks connection state	3, 4, 5
NGFW	App control, DPI, threat detection	Up to 7
iptables	Linux-based firewall rules engine	3, 4 (with rule granularity)